

Assignment 1:

P7.1 - Distance from Point to Function

Analytical Approach

$$y = x^2 + 5 \quad \text{Points} = \{(0,0), (-4,0), (-8,0), (2,0), (6,0)\}$$

$$d = \sqrt{(x - x_0)^2 + (y - y_0)^2}$$

$$D' = 2(x - x_0) + 2(x^2 + 5 - y_0)(2x)$$

$$= 2(x - x_0) + 4x^3 + 20x - 4xy_0$$

$$= 2x - 2x_0 + 4x^3 + 20x - 4xy_0$$

$$= 4x^3 + 2x(11 + 2y_0) - 2x_0$$

$$D' = 0 \quad \text{to find crit. points:}$$

$$0 = 2x^3 + x(11 + 2y_0) - x_0$$

$$\text{Let } x_0, y_0 = (0,0):$$

$$0 = 2x^3 + 11x$$

$$0 = 2x(x^2 + 5.5) \quad \text{not real sol'n}$$

$$x = 0$$

$$\text{When } x = 0, y = 5$$

$$d = \sqrt{0^2 + 5^2} = 5 //$$

$$\text{Let } x_0, y_0 = (-8,0):$$

$$0 = 2x^3 + x(11 + 2(0)) + 8$$

$$0 = 2x^3 + 11x + 8$$

$$x = -0.672, 0.336 \pm 2.416i$$

$$\text{When } x = -0.672, y = 5.452$$

$$d = \sqrt{(-0.672 + 8)^2 + 5.452^2}$$

$$= 9.134 //$$

$$\text{Let } x_0, y_0 = (6,0):$$

$$0 = 2x^3 + x(11 + 2(0)) - 6$$

$$0 = 2x^3 + 11x - 6$$

$$\text{When } x = 0.520, y = 5.27$$

$$x = 0.520, -0.260 \pm 2.39i \quad d = \sqrt{(0.52 - 6)^2 + 5.27^2} = 7.602 //$$

$$\text{Let } x_0, y_0 = (-4,0):$$

$$0 = 2x^3 + x(11 + 2(0)) + 4$$

$$0 = 2x^3 + 11x + 4$$

This is a complex equation that will not have any real integer roots. Solving using a calculator:

$$x = -0.355, 0.178 \pm 2.365i$$

$$\text{When } x = -0.355, y = 5.126$$

$$d = \sqrt{(-0.355 + 4)^2 + 5.126^2} = 6.290$$

$$\text{Let } x_0, y_0 = (2,0):$$

$$0 = 2x^3 + x(11 + 2(0)) - 2$$

$$0 = 2x^3 + 11x - 2$$

$$x = 0.181, -0.09 \pm 2.35i$$

$$\text{When } x = 0.181, y = 5.033$$

$$d = \sqrt{(0.181 - 2)^2 + 5.033^2} = 5.352$$